Operational Memo -- Scenario C (D5): Governance Across 14 / 6 / 0 Crew Regimes

To: SDV Program Office & Vessel Operations Command

From: Principal Systems Architect, sdv-resource-governor

Subject: Deterministic Resource, Deck-Safety, Implicit Timing & Autonomy Rules Under Crew Regimes 14 / 6 / 0

1. Executive Summary

A Software-Defined Vessel (SDV) cannot treat human crew availability (14 / 6 / 0) as a mere scheduling annotation: crew regime directly governs which physical payload transitions are admissible, how long manual confirmation and flight-deck clearing take (30 s vs. 120 s confirmation; 240 s vs. 600 s deck clearing; prohibited if occupied under 0), whether at-sea vehicle replacement (A' window) is physically safe, and how fault recovery escalates when telemetry degrades or a payload overconsumes. In sdv-resource-governor, the 14 / 6 / 0 crew dial, flight-deck clearance (deck_clear), and port-only installation constraints are enforced as hard operational guards inside scheduling.py, admission.py, degradation.py, and replay.py.

2. Quantitative Regime Comparison (14 / 6 / 0 Crew Dial: FULL_CREW_14 vs. REDUCED_CREW_6 vs. UNCREWED_0)

Operational Dimension	Regime 1: 14 Crew (FULL_CREW_14)	Regime 2: 6 Crew (REDUCED_CREW_6)	Regime 3: 0 Crew (UNCREWED_0)
Bridge & Engine Watch	5 personnel	4 personnel	0 (Remote Shore Control + Onboard Autonomy)
Available Deck/Payload Crew	9 personnel	2 personnel	0 personnel
Max Concurrent Manual Deck Teams	2 teams (for non-conflicting manual tasks)	1 team (strictly serialized)	0 manual deck evolutions
Non-Urgent Operator Confirmation (Trigger -> Effect -> Confirmation)	30 s nominal (15-60 s local bridge watch confirmation)	120 s nominal (60-180 s watch handover confirmation)	0 s human confirmation: automated software mode transitions permitted; physical/manual deck evolutions strictly PROHIBITED
Flight-Deck Clearing Latency (when deck_clear=false)	240 s (4 min with dedicated deck crew)	600 s (10 min serialized single-team deck clearing)	PROHIBITED if deck is occupied (0 crew cannot manually clear deck equipment at sea)
G4 UAV Launch / Recovery (5-12 kn into wind, wind <= 25 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; 30 s confirmation; 240 s deck clear if needed)	PERMITTED (2 deck crew; 120 s confirmation; 600 s deck clear if needed)	FORBIDDEN for manual deck handling; if deck occupied, recovery is PROHIBITED (0 crew)
G2 Winch / Probe Deployment (0-5 kn, wind <= 20 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; cannot overlap G4 flight ops)	PERMITTED (2 deck crew; serialized)	FORBIDDEN (requires umbilical/winch deck handling)
Simultaneous G2 Winch + G4 Flight Ops	FORBIDDEN (Annex A.5 requires flight deck clear during G4 ops; 0-5 kn vs 5-12 kn only touch at 5.0 kn with conflicting heading vs station-keeping)	FORBIDDEN (insufficient crew + Annex A.5 deck clearance + conflicting heading/station-keeping profiles)	FORBIDDEN (0 crew)
At-Sea Vehicle Replacement A' (4-8 kn, wind <= 20 kn, SS <= 3; G4: 3 crew/30m, G3: 4 crew/45m)	PERMITTED (available_crew=9 >= 4 for G3 and >= 3 for G4; earliest window T+1260 s)	BLOCKED (available_crew=2 < 3 for G4 and < 4 for G3 safe replacement at sea)	FORBIDDEN (0 crew; all vehicle replacements deferred to port)
At-Sea Bay/Mast or >250 kg Module Install (G5, G6 420 kg in bay2/bay1_a)	FORBIDDEN AT SEA (Annex A.5 bay/mast installs are port-only; Annex A.4 limits at-sea packages to <= 250 kg)	FORBIDDEN AT SEA (port-only bay/mast + >250 kg + available_crew=2 < 4)	FORBIDDEN AT SEA (CRW-02: zero physical installations at sea)

3. Concrete Implicit Rule (Trigger -> Effect -> Confirmation) & Physical Interlocks Enforced by the Solver

* 1. Concrete Implicit Rule (Trigger -> Effect -> Confirmation Across 14 / 6 / 0):
  * Trigger: Any operational event requiring deck state change, mode escalation, or recovery window preparation (e.g., G4 18% battery recovery alert or deck_clear=false prior to flight operations).
  * Effect: Urgent safety/protection actions (STOP, HOLD, SECURE <= 200 ms, G3 ACOUSTIC_HOLD, or G5 overload step-down) execute autonomously without waiting for human confirmation across all 14 / 6 / 0 regimes. For non-urgent manual evolutions and physical deck preparation, crew regime dictates both confirmation latency (30 s under 14 crew vs. 120 s under 6 crew vs. automated software-only / physical evolution prohibited under 0 crew) and flight-deck clearing time (240 s under 14 crew vs. 600 s under 6 crew vs. strictly prohibited if occupied under 0 crew).
  * Confirmation & Precondition Gate: In Scenario B (14 crew default), G4 recovery adds confirmation + acoustic hold handshake (15 s nominal, [10, 60] s valid window) + vessel turn/stabilization (90 s) + approach/landing (240 s). If switched to 6 crew with an occupied deck (600 s deck clearing + 120 s confirmation), total preparation exceeds the 720 s (12 min) battery emergency window (480 + 600 + 120 + 90 + 240 = 1,530 s > 1,200 s), causing the solver to flag the plan as infeasible unless the deck is pre-cleared! Under 0 crew, any evolution requiring manual deck clearing (deck_clear=false) or manual vehicle handling is immediately rejected (PROHIBITED).
* 2. Why G2 and G4 Cannot Execute Simultaneous Deck Operations Even With 14 Crew (Annex A.5):
  * While 14 crew provides 9 deck/payload personnel (enough to staff two manual handling teams for non-conflicting tasks), G2 ROV/winch deployment and G4 UAV launch/recovery are physically incompatible: (a) Annex A.5 mandates that "Launch or recovery of a drone requires the flight deck to be clear of other ongoing deck operations" (state.deck_clear == True), and (b) G2 requires low-speed station-keeping (0-5 kn) whereas G4 requires 5-12 kn with vessel heading aligned within +/- 15 deg of the wind (their speed ranges only touch at the single 5.0 kn boundary, where deck_clear and heading/winch-handling rules still strictly forbid overlap).
* 3. Why G6 (420 kg, bay2/bay1_a) Is Strictly Port-Only Across All 14 / 6 / 0 Crew Regimes (Annex A.4 & A.5):
  * Annex A.5 specifies that payload installation in Bays 1-2 and Mast Positions 1-3 is a port-only operation, and Annex A.4 restricts at-sea A' package installation to light deck packages <= 250 kg in Sea State <= 3, wind <= 20 kn, and 4-8 kn. Because G6 weighs 420 kg > 250 kg and installs into bay2 / bay1_a, G6 has port_only: true and is rejected (NO / NOT_NOW) for at-sea installation under 14, 6, and 0 crew alike.
* 4. Acoustic Survey Deck-Machinery Lockout (G3 forbids_deck_machinery_while_active=True):
  * Across 14 / 6 / 0 crew regimes, while G3 is in SURVEY_QUIET (1 kW, 2-6 kn), hydraulic A-frames, winches (G2), and flight-deck handling gear (G4) are locked out to prevent acoustic corruption of the seabed survey sonar until ACOUSTIC_HOLD_CONFIRMED ACK is verified.

4. Fault Governance & Component-Split Overconsumption Under 14 / 6 / 0 Regimes

* Urgent Safety Commands (STOP, HOLD, SECURE <= 200 ms, SAF-01): Routed strictly over the dual 10 Gbps local shipboard LAN (< 5 ms transport latency), never over 600-800 ms SATCOM. Measured software dispatch latency on both MacBook Pro (arm64) and the live Arduino Ventuno Q (training06, aarch64) is < 1 ms (<< 200 ms deadline).
* Guest Overconsumption (+20% Component-Split Accounting, RUN-04, ?6.5): Because none of G1-G5 is safety-critical by design, if G5 (45.0 kW nominal = 43.8 kW container chilled-water + 1.2 kW roof-antenna air) overconsumes by +20% to 54.0 kW electrical (+9.0 kW surge absorbed by the 3,000 kg compute container -> 52.8 kW chilled-water + 1.2 kW roof air), the chilled-water ledger immediately reflects the +9.0 kW thermal spike and the governor autonomously throttles G5 to DEGRADED_70PCT_LOSS (8.0 kW electrical = 7.5 kW water + 0.5 kW roof air, retaining 30% capability) while protecting active uninterruptible/recovery tasks (G3 recharge, G4 flight/recovery).
* Lost or Stale Telemetry Feeds (RUN-03): If seawater_temp_c exceeds its TTL (STALE / UNAVAILABLE), the governor never extrapolates optimistic cooling; it clamps to the conservative 30 deg C (120 kW nominal / 72 kW degraded) thermal bound until sensor freshness is restored.